

QUESTIONS 10 / 10 completed

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion.

Sample Input: Enter a number- 6

Sample Output: Factorial of 6 is 720

"

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 #include<time.h>
3 int calls=0;
4 int fib(int n)
5 {
6     calls++;
7     if(n<=1)
8         return n;
9     return fib(n-1)+ fib(n-2);
10 }
11 int main()
12 {
13     int n,i,a=0,b=1,c;
14     clock_t start,end;
15     scanf("%d",&n);
16     printf("iterative: ");
17     start=clock();
18     for(i=0;i<n;i++)
19     {
20         if(i--0)
21             printf("%0 ");
```

OUTPUT

```
iterative: 0 1 1 2 3 5 8
recursive: 0 1 1 2 3 5 8
recursive calls:59
iterative time 0.000 ms
recursive time:0.000 ms
```

INPUT

7

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```
22     else if(i==1)
23         printf("1 ");
24     else{
25         c=a+b;
26         printf("%d ",c);
27         a=b;
28         b=c;
29     }
30 }
31 end=clock();
32 printf("\n");
33 printf("recursive: ");
34 calls=0;
35 start=clock();
36 for(i=0;i<n;i++)
37     printf("%d ",fib(i));
38 end=clock();
39 printf("\nrecursive calls:%d",calls);
40 printf("\niterative time %.3f ms",0.0);
41 printf("\nrecursive time: %.3f ms", (double)(end-start)*1000/CLOCKS_PER_SEC);
42 return 0;
```

OUTPUT

```
iterative: 0 1 1 2 3 5 8
recursive: 0 1 1 2 3 5 8
recursive calls:59
iterative time 0.000 ms
recursive time:0.000 ms
```

INPUT

7